

MEA OCPP 1.6 Compliance Test Report

Section 7: Abnormal Operation Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 05:20 UTC

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1 Test Configuration

Device Under Test Vector vSECC.single Board
Device IP 192.168.1.166
OCPP Version 1.6 (JSON)
CP ID rddQC4000001
CSMS wss://ocpp.measandbox.com:2930
Connection Direct TLS (Security Profile 1, no proxy)
Test Date 2026-06-11 05:20 UTC
Results file tex/vsecc_section7_results.json

Test Architecture

The vSECC connects directly to the MEA CSMS at `wss://ocpp.measandbox.com:2930/EV/Srv/JSON/1.6/rddQC4000001` over TLS with no intermediate proxy. OCPP traffic is observed via the vSECC's `GET /api/logging/files/ocpplib.log` REST endpoint (TRACE-level log, raw » sent / « received frames) and via MQTT at 192.168.1.166:1883 (topic `vsecc/#`). The test PC NAT-routes internet access for the vSECC subnet (192.168.1.x via `enp3s0`).

2 Section 7 Results

Summary: 9 PASS 0 FAIL 45 WARN 0 SKIP (54 total)

Item	Test Description	Detail / Evidence	Result
7.1.1	StatusNotification Available (charger idle, no EV)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.1.2	RemoteStartTransaction (EV not connected) HTTP 200	HTTP 200 <i>vSECC should respond Rejected when no EV present</i>	PASS
""			
7.2.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.2.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.2.3	RemoteStartTransaction (first) HTTP 200	HTTP 200	PASS
""			
7.2.4	RemoteStartTransaction (second, concurrent) HTTP 200	HTTP 200 <i>CSMS likely blocks duplicate; charger-level Rejected not observable without proxy</i>	PASS
""			
7.2.5	StartTransaction (session started after first RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.2.6	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.2.7	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN
7.2.8	RemoteStopTransaction HTTP 200	HTTP 200	PASS
""			
7.2.9	StopTransaction (session ended after RemoteStop)	session_state idle/finished not observed on MQTT in 25 s	WARN

Item	Test Description	Detail / Evidence	Result
7.2.10	StatusNotification Finishing	Finishing not observed on MQTT in 15 s <i>Some implementations skip Finishing and go directly to Available</i>	WARN
7.2.11	StatusNotification Available (after stop / unplug)	Available not observed on MQTT in 30 s unplug EV	WARN
7.3.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.3.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.3.3	Authorize (invalid RFID tag) Rejected	Not triggerable via API vSECC reads RFID directly from hardware <i>Requires physical RFID scan on vSECC hardware of unknown/blocked card</i>	WARN
7.3.4	Authorize (valid RFID tag) Accepted	Not triggerable via API vSECC reads RFID directly from hardware <i>Requires physical RFID scan on vSECC hardware of authorized card after rejected card</i>	WARN
7.3.5	StartTransaction (after valid RFID)	Depends on 7.3.3/7.3.4 (physical RFID scan) <i>Session starts only after valid RFID tap on vSECC hardware</i>	WARN
7.3.6	StatusNotification Charging (after valid auth)	Depends on 7.3.5 (StartTransaction) <i>Requires completed Authorize sequence via physical RFID hardware</i>	WARN
7.3.7	MeterValues (during charging)	Depends on 7.3.6 (Charging state) <i>Requires completed Authorize sequence via physical RFID hardware</i>	WARN
7.3.8	RemoteStopTransaction	Depends on 7.3.5 (StartTransaction) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.9	StopTransaction (after RemoteStop)	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.10	StatusNotification Finishing	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.3.11	StatusNotification Available (after stop / unplug)	Depends on 7.3.8 (RemoteStop) <i>Requires active session started via physical RFID hardware</i>	WARN
7.4.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.4.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.4.3	RemoteStartTransaction HTTP 200	HTTP 200	PASS
""			
7.4.4	StartTransaction (session started after RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.4.5	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.4.6	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN

Item	Test Description	Detail / Evidence	Result
7.4.7	StopTransaction (reason=EmergencyStop)	Triggered by physical emergency stop button on vSECC hardware <i>Requires physical emergency stop button press on the vSECC unit</i>	WARN
7.4.8	StatusNotification Faulted (after EmergencyStop)	Faulted status not observed on MQTT (E-stop not pressed) <i>Requires physical emergency stop button press on the vSECC unit</i>	WARN
7.4.9	StatusNotification Available (E-stop cleared)	Depends on 7.4.7 (physical E-stop press and release) <i>E-stop must be physically released to restore normal operation</i>	WARN
7.4.10	StatusNotification Available (EV unplug after E-stop recovery)	Depends on 7.4.7 (physical E-stop press and release) <i>EV must be unplugged after E-stop recovery to reach Available</i>	WARN
7.5.1	StatusNotification Available (before door open)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.5.2	StatusNotification Faulted (reason=DoorOpen)	Triggered by opening physical door/panel on vSECC hardware <i>Requires physical door open on vSECC hardware; not triggerable via API</i>	WARN
7.5.3	StatusNotification Available (after door closed)	Depends on 7.5.2 (physical door open and close) <i>Door must be physically closed to restore normal operation</i>	WARN
7.6.1	StatusNotification Available (before EV plug)	Available not observed on MQTT in 10 s <i>Check CSMS event log; charger may already be available</i>	WARN
7.6.2	StatusNotification Preparing (EV plug)	EV_WAIT_SEC=0 skipped <i>Connect physical EV to connector and rerun</i>	WARN
7.6.3	RemoteStartTransaction HTTP 200	HTTP 200	PASS
""			
7.6.4	StartTransaction (session started after RemoteStart)	charging_session_state not observed on MQTT in 30 s	WARN
7.6.5	StatusNotification Charging	Charging status not observed on MQTT in 20 s	WARN
7.6.6	MeterValues (during charging)	MeterValues not observed on MQTT in 15 s	WARN
7.6.7	StopTransaction (PowerLoss) simulated via Hard Reset HTTP 200	HTTP 200 <i>Simulated via Hard Reset; physical power loss not testable in this environment</i>	PASS
""			
7.6.8	BootNotification Accepted (vSECC reconnected after reboot)	CSMS reconnection confirmed after Hard Reset	PASS
7.6.9	StatusNotification Preparing (EV still connected after reboot)	Preparing not observed on MQTT in 20 s after reboot <i>EV must remain connected through the reboot sequence</i>	WARN
7.6.10	StatusNotification Available (EV unplug after reboot)	Available not observed on MQTT in 30 s unplug EV	WARN

Item	Test Description	Detail / Evidence	Result
7.7.1	SendLocalList (version=1, RFID_TEST=Accepted) HTTP 200	HTTP 200	PASS
""			
7.7.2	StatusNotification Available (charger online before offline test)	Offline sequence not observable via MQTT/REST without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.3	StatusNotification Preparing (EV plug while charger offline)	Offline start requires disconnecting vSECC from CSMS network <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.4	Authorize (local list lookup while offline) Accepted	vSECC must authorize from local list while CSMS unreachable <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.5	StartTransaction (offline, buffered by vSECC)	vSECC buffers transaction in firmware; not observable without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.6	StopTransaction (offline, buffered by vSECC)	vSECC buffers stop in firmware; delivered when CSMS reconnects <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN
7.7.7	CSMS reconnect buffered Start/StopTransaction flushed	Reconnect flushing is firmware-internal; not observable without proxy <i>Offline buffering not observable via MQTT/REST without proxy</i>	WARN

3 Notes

Remote-controllable flows (7.2, 7.6)

RemoteStart/RemoteStop and Reset commands are verifiable via MEA REST API and ocpplib.log.

Physical-hardware items (7.3, 7.4.7–7.4.10, 7.5)

Card-swap, emergency stop, door-open switch, and power-loss scenarios require physical interaction with the hardware. These are permanently recorded as WARN with a remark indicating the required operator action.

Offline buffering (7.7)

Offline transaction buffering cannot be verified without direct OCPP frame capture from the reconnection burst. Recorded as WARN pending further tooling.